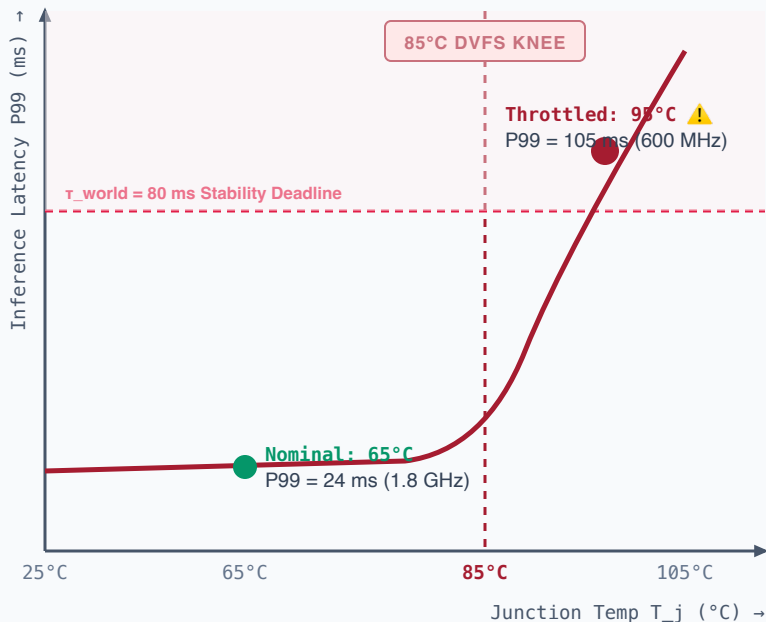


THERMAL DERATING CLIFF & THE RE-PLACEMENT RIPPLE EFFECT

Silicon DVFS throttling crossing the 85°C junction knee: latency tripling, deadline breach, and stopping distance expansion

SILICON JUNCTION TEMPERATURE VS. INFERENCE TAIL LATENCY



PHYSICAL RE-PLACEMENT RIPPLE EFFECT

KINETIC CONSEQUENCE: STOPPING DISTANCE COLLAPSE

$$d_{\text{stop}} = v \cdot (t_{\text{latency}} + t_{\text{brake}}) + v^2 / (2 a_{\text{max}})$$

NOMINAL ($T_j = 65^\circ\text{C}$)

$$d_{\text{stop}} = 0.35 \text{ m (Safe)}$$

THROTTLED ($T_j = 95^\circ\text{C}$)

$$d_{\text{stop}} = 0.82 \text{ m } \times \text{COLLISION}$$

- Thermal throttle causes **134% increase in stopping distance**, breaching clearance.

ADAPTIVE RE-PLACEMENT CODESIGN STRATEGIES

- Dynamic Model Switching:** Shift from heavy ViT to quantized INT4 MobileNet to stay strictly within onboard thermal power envelope.
- Adaptive Velocity Scaling:** Reduce robot max velocity v_{max} proportionally to measured silicon junction temperature.
- Cloud Offloading with Local Leases:** Offload System 2 deliberation while keeping System 1 reflexes anchored to bare-metal MCU.